BUILD YOUR FIRST AI AGENT

Assessment Plan

A hands-on introduction to Agentic AI using Python and LLM APIs

Technical Trainer Assessment

Course duration: 3 days × 3 hours per day | Beginner level

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1. Assessment Purpose

The assessment checks conceptual understanding, practical implementation and the ability to explain how an AI agent works.

2. Assessment Structure

Assessment component	Marks	Weight
Five-question quiz	10	20%
Hands-on assignment	30	60%
Practical demonstration and explanation	10	20%
TOTAL	50	100%

The quiz contains five questions worth 2 marks each. The assignment is worth 30 marks. The practical demonstration is worth 10 marks.

3. Knowledge Check — Five Quiz Questions

Each question carries 2 marks. Select one correct answer for each question.

Q1. What is the primary difference between a traditional chatbot and a tool-using AI agent?

A. A chatbot always uses databases

B. An agent can select and execute tools to accomplish tasks

C. An agent only generates predefined responses

D. A chatbot cannot understand natural language

Correct answer: *B — An agent can select and execute tools to accomplish tasks.*

Q2. Which component is primarily responsible for selecting which tool to call?

A. CSS

B. HTML

C. LLM

D. Database

Correct answer: *C — The LLM selects or requests an appropriate tool based on the task and available tool descriptions.*

Q3. What is the correct sequence in a basic agent loop?

A. Act → Plan → Observe

B. Observe → Act → Plan

C. Plan → Act → Observe

D. Plan → Observe → Delete

Correct answer: *C — Plan → Act → Observe.*

Q4. What is function calling?

A. Executing every function automatically

B. A mechanism where the model requests a structured function call that the application executes

C. Training a model using Python functions

D. A method for storing images

Correct answer: *B — The model requests a structured function call; the application executes it and returns the result.*

Q5. Which is an appropriate way to prevent an agent from endlessly calling tools?

A. Remove all system prompts

B. Increase the number of tools

C. Add iteration limits and stopping conditions

D. Allow unrestricted code execution

Correct answer: *C — Iteration limits and stopping conditions control the agent loop.*

Quiz Marking Scheme

Response	**Marks**
Correct answer	2
Incorrect answer	0

Unanswered	0
Negative marking	None
Maximum score	10

Suggested Score Interpretation

Score	**Performance**
9–10	Excellent understanding
7–8	Good understanding
5–6	Basic understanding
0–4	Requires revision

4. Hands-on Assignment — 30 Marks

Assignment title: Build a Smart Student Assistant Agent.

Problem Statement

Build a Python-based AI agent that helps students manage academic activities. The agent should understand natural-language requests and select the appropriate tools to complete the requested operation.

Required Tools

Tool / Function	**Purpose**
add_task(title, subject, deadline)	Add an academic task.
list_tasks()	Display pending tasks.
calculate_percentage(obtained, total)	Calculate marks percentage.
search_notes(query)	Search a small predefined collection of study notes.

Example Interaction

Student: “Add a Python assignment due tomorrow.”

Agent action: Requests add_task(title='Python Assignment', subject='Python', deadline='Tomorrow').

Agent response: “Your Python assignment has been added.”

Student: “I scored 72 out of 90. What's my percentage?”

Agent response: “Your percentage is 80%.”

Assignment Requirements

- Connect the application to an LLM API.
- Define at least three functional tools.
- Implement a tool-calling loop.
- Use structured tool arguments.
- Handle invalid inputs and tool errors.
- Demonstrate at least five successful test scenarios.
- Submit source code and a short README.

Assignment Marking Rubric

Evaluation criterion	What is assessed	Marks
LLM API integration	Correct API connection and request/response handling	5
Tool/function implementation	Working functions, appropriate parameters and expected results	8
Agent loop and tool selection	Correct tool selection, execution and result hand-off	6
Input validation and error handling	Validation of inputs and handling of tool/API errors	4
Testing and demonstration	At least five test scenarios and a clear demonstration	4

Code quality and documentation	Readable structure, source code and short README	3
TOTAL		30

5. Practical Demonstration — 10 Marks

Students demonstrate their application and explain the agent architecture, tool selection, execution and handling of an unsuccessful operation.

Criterion	**Evidence expected**	**Marks**
Explains the agent architecture	Identifies the LLM, tools, application and data flow	3
Demonstrates appropriate tool selection	Shows the agent selecting a suitable tool for a user request	3
Explains execution and observation	Explains how Python executes the tool and returns its result	2
Handles a failure scenario	Shows or explains validation, tool error handling or safe stopping	2
TOTAL		10

6. Overall Marking and Competency Guide

Component	**Maximum marks**
Quiz	10
Hands-on assignment	30
Practical demonstration	10
TOTAL	50

Suggested competency threshold: A student should achieve at least 25 out of 50 to demonstrate basic competency. Use the assignment and practical demonstration to identify whether the student can independently implement and explain an agent.

7. Assessment Administration Notes

- Explain the assessment requirements and marking criteria before students begin.
- Allow students to use the course examples as a starting point, but assess their own implementation and explanation.
- Check that tools produce correct results rather than awarding marks only for natural-language responses.
- Include at least one invalid-input or tool-failure test.
- Award marks according to demonstrated functionality and understanding.
- Use quiz results to identify topics that may need revision.

8. Alignment Between Learning Outcomes and Assessment

Learning outcome	**Where it is assessed**
Explain AI, LLMs and Agentic AI	Quiz; practical explanation
Understand LLM API communication	Assignment: API integration
Differentiate chatbot and agent	Quiz; practical explanation
Understand function calling and tools	Quiz; assignment: tool implementation
Implement Plan → Act → Observe	Quiz; assignment: agent loop
Build a Python agent	Hands-on assignment and demonstration
Connect tools and use structured arguments	Assignment: required tools and tool schemas
Identify failures and security risks	Quiz; validation/error handling; practical demonstration